FastIdentity

Platform guide

First-party, consent-first identity resolution for B2C websites

Pixel · Identity graph · Abandonment flows · GDPR / PECR compliance

identity.fastsme.com · November 2026

1. Executive summary

What it is. FastIdentity is a self-hosted identity layer for consumer websites. A small consent-aware pixel and a server API collect first-party events. An identity graph then stitches anonymous browsing to known people, and an activation layer triggers abandonment emails through the ESP you already use.

Why it matters. Most visitors to a high-consideration site (medical travel, finance, property) browse anonymously for weeks before enquiring. Third-party cookies and ad-tech IDs are gone or unreliable. First-party identity, built on consent, is now the only durable way to recognise a returning visitor and follow up at the right moment.

How it works.

- **Deterministic first.** Email, phone, user ID and CRM ID merge profiles with certainty, and every merge can be undone.
- **Probabilistic second, and only with consent.** A Fellegi–Sunter model scores device and network signals, which are stored as keyed hashes and never as raw fingerprints. Matches above 97% merge automatically; matches between 70% and 97% go to a human review queue. An ambiguity guard never auto-merges an anonymous visitor who resembles two known people.
- **Devices and households.** Each person's devices are listed: desktop, phone, tablet or app, identified by an explicit device ID when available and by a fingerprint otherwise. Shared devices are detected. Households are opt-in: profiles that repeatedly share a home IP are grouped, never merged.
- **Compliance built in.** It keeps a consent ledger per purpose and legal basis, maps CookieYes, OneTrust and Google Consent Mode v2 consent automatically, and handles PECR soft opt-in. Suppression is hashed, every action is audit-logged, and right-to-erasure and data-export (DSAR) requests are served through an API.

Where we are. The platform is live at `identity.fastsme.com` on the FastSME Coolify stack: UK data region, PostgreSQL, CI/CD on every push to `main`, and 56 automated tests. It runs as multi-tenant SaaS: each customer gets a workspace with its own members, roles and secret API keys. The first customer integration is **MyMedicalGateway**; see Section 8.

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An animated walkthrough of every screen is in `docs/demo/fastidentity-walkthrough.gif`.

3. Comparables — EEA-native identity & CDP vendors

Vendor	HQ	What they resolve	Model	Relevance to us
Zeotap	Berlin, DE	First-party CDP with its own identity graph; strong on consent and EU hosting	SaaS, enterprise	Closest positioning (“privacy-first CDP”), aimed at telco and retail enterprises
Utiq	Brussels, BE	Telco-network IDs (a JV of Deutsche Telekom, Orange, Telefónica and Vodafone); per-site opt-in	Network-level ID for publishers and advertisers	Ad-tech addressability, not CRM-grade stitching
Bloomreach (ex-Exponea)	Bratislava, SK / US	Engagement CDP: profile merging on customer IDs, plus email and SMS	SaaS suite	Bundles ESP and CDP; high switching cost
Synerise	Kraków, PL	AI-driven CDP with identity stitching and real-time segments	SaaS suite	Retail and banking focus in CEE
Meiro	Prague, CZ	Composable CDP with deterministic stitching; can be self-hosted	Software + services	Similar “own your data” pitch; deployments are heavier
Commanders Act	Paris, FR	Tag manager + CMP + CDP in one stack	SaaS suite	Strong on French consent rules (CNIL)
Snowplow	London, UK	Behavioural event pipeline with identity stitching in the warehouse	Open core / SaaS	Pipeline, not activation; needs a data team
ID5	London, UK	Universal ad ID built from hashed email and probabilistic signals	Ad-tech ID	Advertising only; no CRM graph

Compiled from vendors’ public positioning as of November 2026; check before using externally. “EEA-native” means headquartered or primarily hosted in the EEA or UK.

3. Comparables — global incumbents operating in the EEA

Vendor	What they resolve	EEA posture	Gap FastIdentity fills
Twilio Segment (Unify)	Deterministic profile stitching across sources	EU workspace region available	Priced per tracked user; probabilistic matching not consent-gated per purpose
mParticle (IDSync, now Rokt)	Rule-based identity strategies for apps and web	EU data residency available	Mobile-first and enterprise-priced
Tealium (AudienceStream)	Visitor stitching on attributes	EU hosting; bundled consent tooling	Tag-manager-centric and heavy to run
Adobe Real-Time CDP / **Salesforce Data Cloud**	Enterprise identity graphs inside each vendor's own suite	EU regions	Suite lock-in; long implementations
Amperity	ML identity resolution for retail	EU instances for large brands	Enterprise only
LiveRamp (RampID, ATS)	Offline + online identity graph for advertising	Large EU footprint	Ad-tech oriented; data leaves your perimeter
Klaviyo / **Braze** / **Customer.io**	Profile merging inside the ESP	EU data centres	Each ESP sees only its own events; no cross-channel graph
Splink (open source, UK MoJ)	Probabilistic record linkage	Runs anywhere	A library rather than a product; FastIdentity can use it as an optional engine

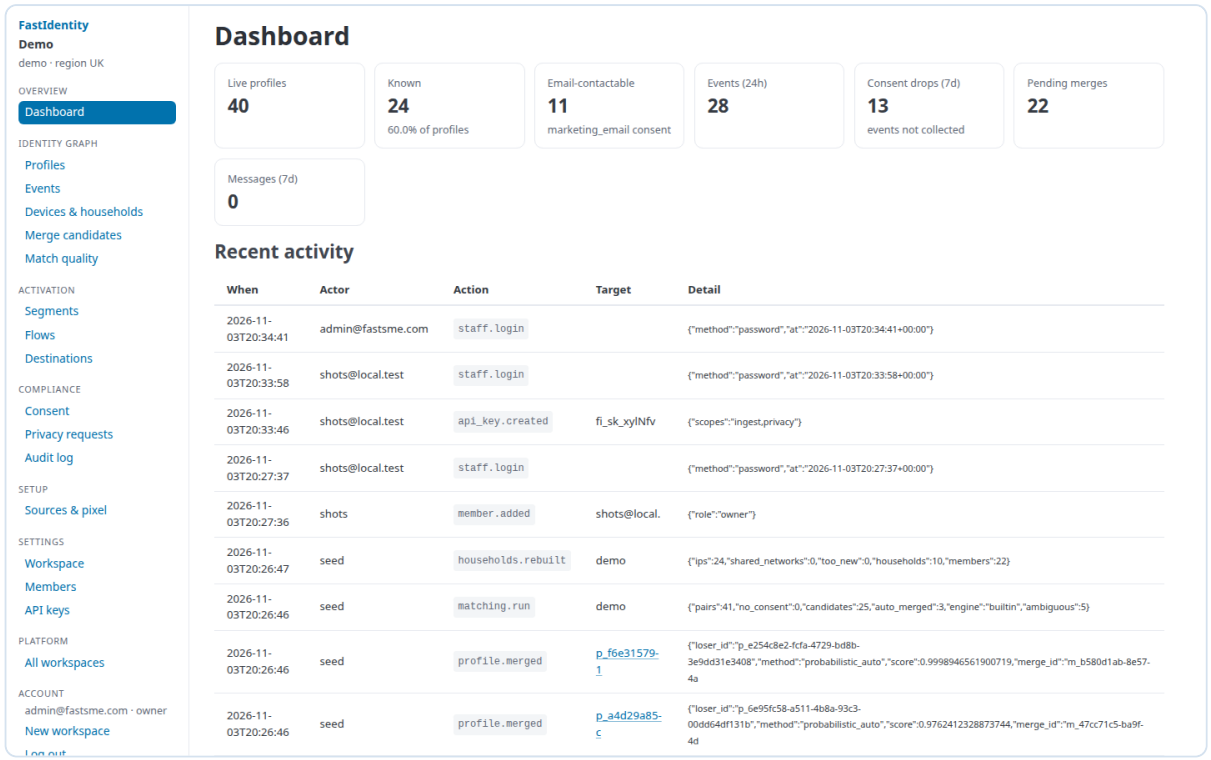
Our wedge. These vendors cover the enterprise suite and the ad-tech ID ends of the market. FastIdentity aims at the gap between them for SMEs in regulated, high-consideration B2C markets:

- it is self-hosted in the UK or EU and sits next to the ESP the business already uses;
- its consent is purpose-level and covers health-adjacent data;
- probabilistic matching is gated on consent and protected by the ambiguity guard;
- it costs a fraction of an enterprise CDP.

4. Overview — dashboard

The **dashboard** is the daily health check.

- **Live / known profiles**: how much of the audience is identified.
- **Email-contactable**: known profiles with valid `marketing_email` consent. This is the real addressable audience.
- **Consent drops**: events discarded because the visitor declined analytics. A rising number usually means a broken CMP mapping.
- **Pending merges**: items waiting in the review queue.
- **Recent activity**: the same entries as the audit log (merges, consent changes, flow runs, logins).

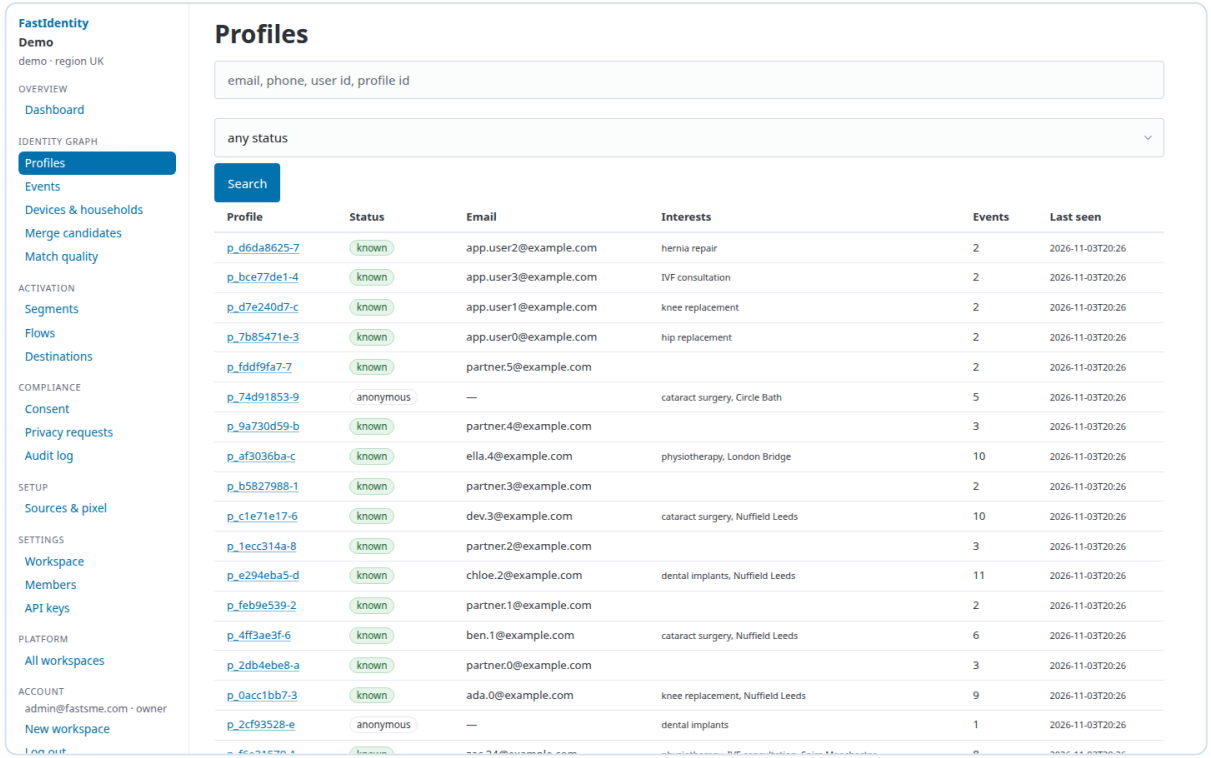


5. Identity graph — profiles

Profiles lists every person the graph knows about. They are either *known* (they have an email, phone or user ID) or *anonymous* (a consenting browser only).

Search by email, phone or ID. A merged profile disappears from the list. Its identifiers move to the surviving profile, and the merge can be **unmerged** later if it was wrong.

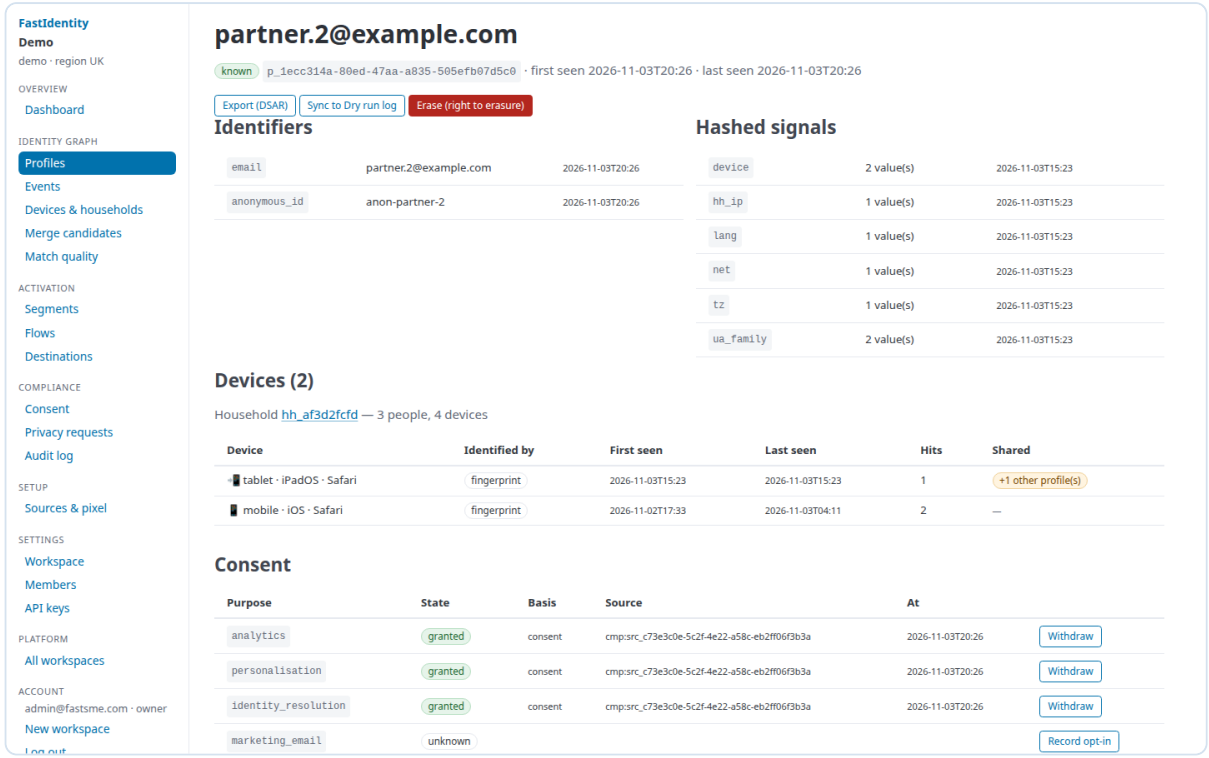
Anonymous profiles that never become known are deleted automatically after 90 days (`ANON_PROFILE_RETENTION_DAYS`).



5. Identity graph — unified profile

One page shows a single person.

- **Identifiers**: every anonymous ID, email, phone and user ID linked to the person, with first-seen dates.
- **Hashed signals**: device, device ID, /24 network, browser family, timezone and language. These are stored only as keyed hashes, and only with `identity_resolution` consent.
- **Devices**: each device the person used (type, OS, browser), whether it was identified by device ID or fingerprint, and whether **another profile shares it**. The person's household, if any, is linked here.
- **Consent**: the current state of each purpose, with its source and legal basis. Staff can record an opt-in or a withdrawal here.
- **Interests**: treatments and hospitals viewed, counted from events.
- **Actions**: export (DSAR), sync to a destination, or **erase** (right to erasure).

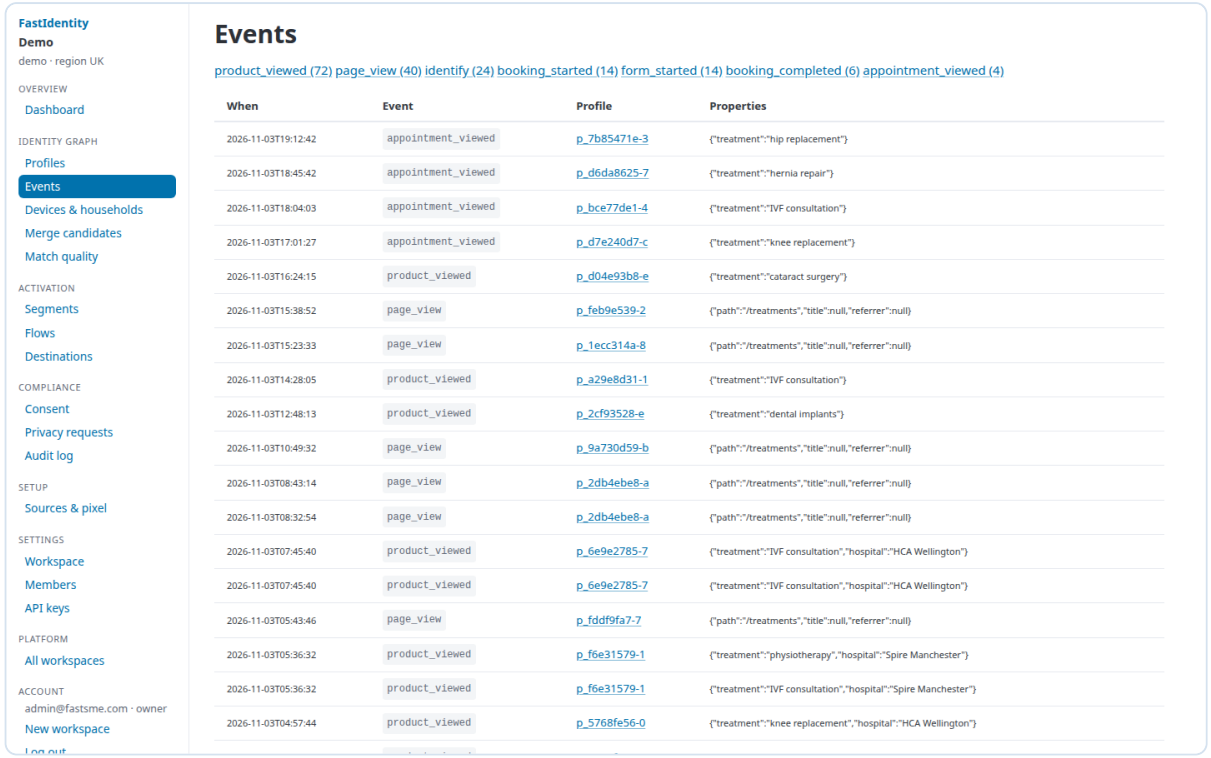


5. Identity graph — event stream

Events is the raw first-party stream: page views, custom events such as `treatment_viewed` or `enquiry_started`, and identify calls.

Each event records the profile it belongs to now and the profile it **originally** arrived on. That is what makes a clean unmerge possible.

PII keys (`email`, `phone`, `name`, …) are stripped from event properties before storage. Identity data goes only through `identify`.

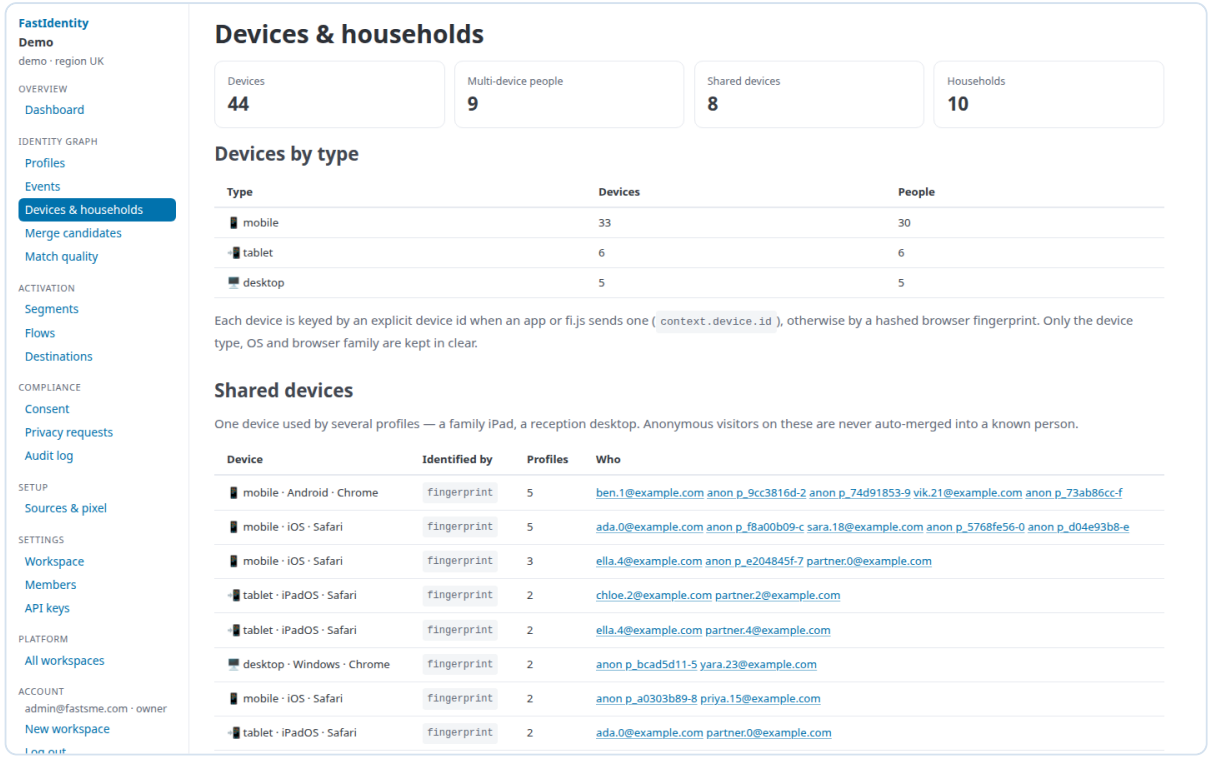


5. Identity graph — devices & households

Devices. Every event with `identity_resolution` consent is tied to a device.

- An **explicit device ID** is used when one is available: the mobile app's vendor ID sent through the backend, or fi.js's first-party ID, which survives logout. Otherwise the device is identified by a hashed **fingerprint**.
- Only the device type, OS and browser family are kept in plain text. Advertising IDs (IDFA, GAID) are ignored.
- **Shared devices**, such as a family iPad or a reception PC, are listed. An anonymous visitor on one is never merged automatically.

Households are opt-in per workspace, switched on by the owner. Profiles are grouped when they share a home IP on at least two days within 30 days. IPs with more than six profiles (offices, mobile carriers, VPNs) are ignored. Switching households off deletes the groups and the hashed IPs.



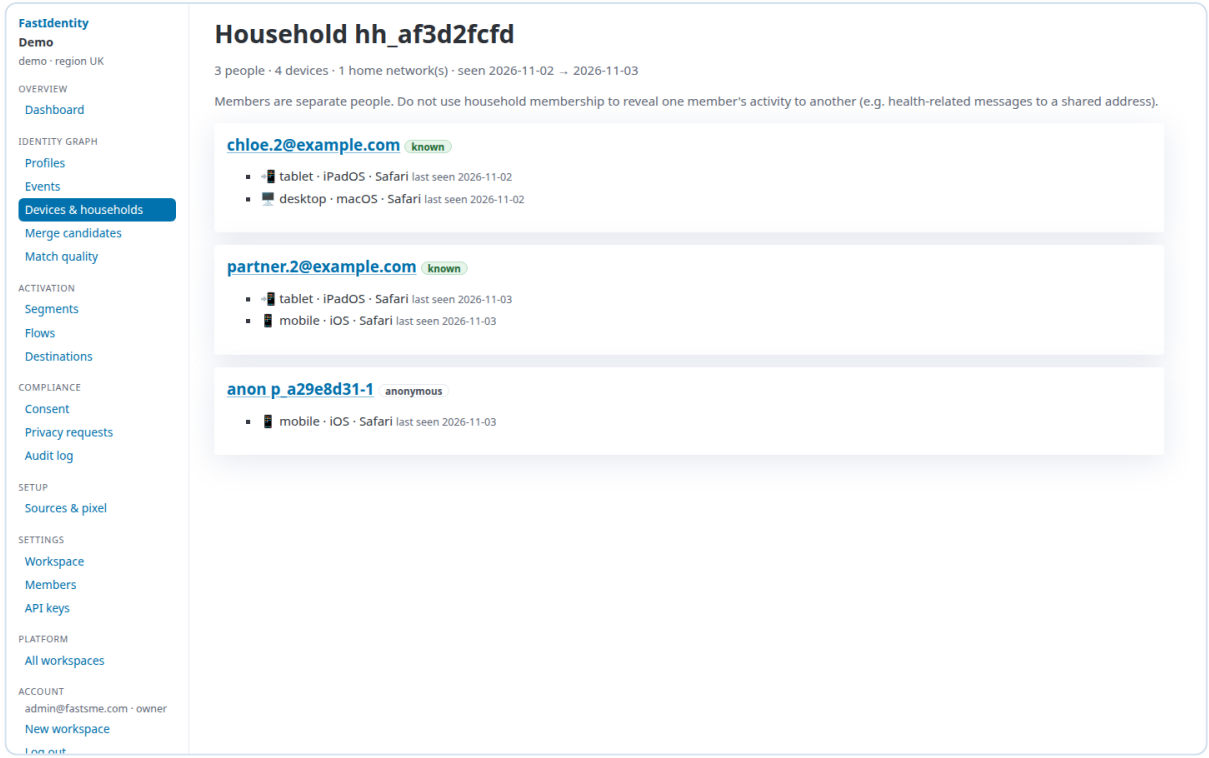
5. Identity graph — one household

A household groups **different people**; it never merges them. This one has two partners and a shared tablet; the third member is an anonymous visit still waiting in the review queue.

Used for

- Explaining merge candidates: “same household” usually means two people, not one.
- Audiences: `household` (min. members) and `device` (mobile, tablet, desktop) segment rules.
- Insight: how many patients research on several devices, and who decides together.

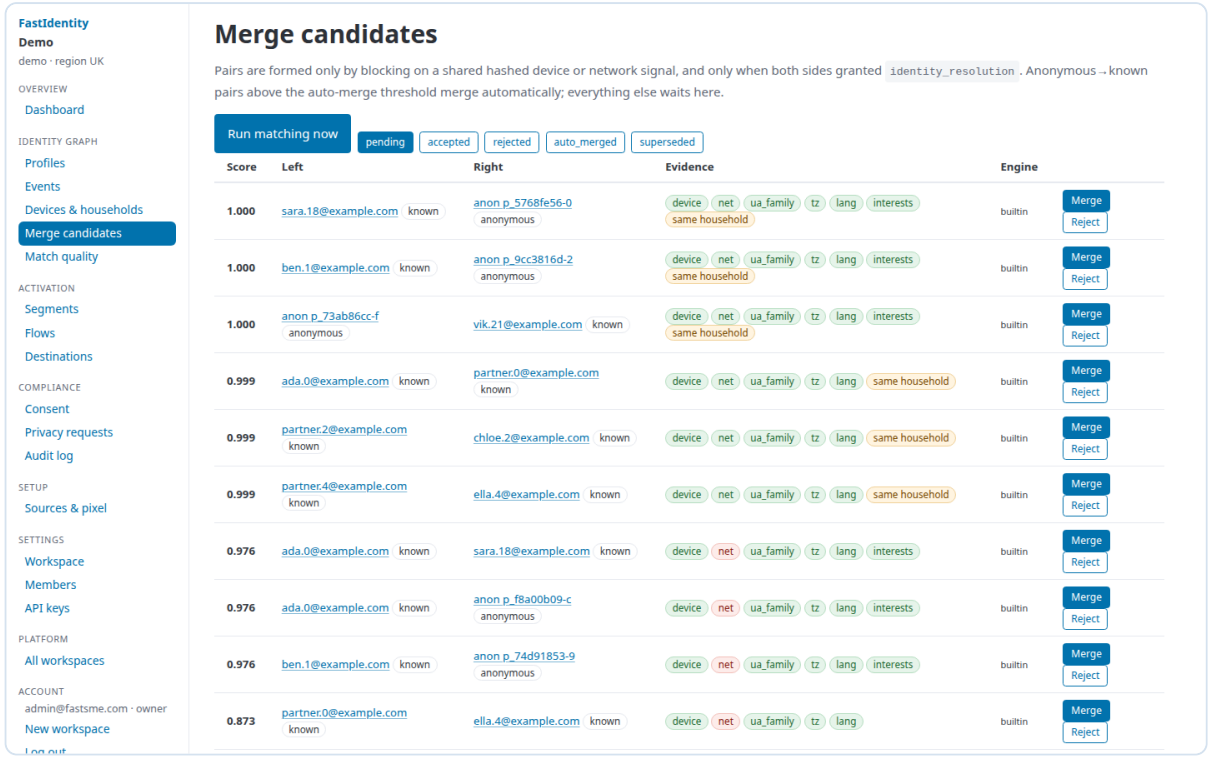
Not used for revealing one member’s activity to another. A data export lists only the person’s own devices and household ID, never the other members. For health sites, leave households off unless a DPIA covers them.



5. Identity graph — merge candidates

The **probabilistic matcher** runs every minute. It compares profiles that share a blocking key (device, device ID or network hash), then scores each pair with Fellegi–Sunter weights.

- **≥ 0.97** merges automatically, but only when every profile involved has `identity_resolution` consent.
- **0.70 – 0.97** goes into this review queue, where staff accept or reject.
- **Ambiguous**: an anonymous visitor who resembles more than one known person is never auto-merged; shared family devices are the usual cause.
- Two *known* people are never merged automatically.
- A **same household** flag marks pairs who live together. These are often two people on one family tablet, not one person.

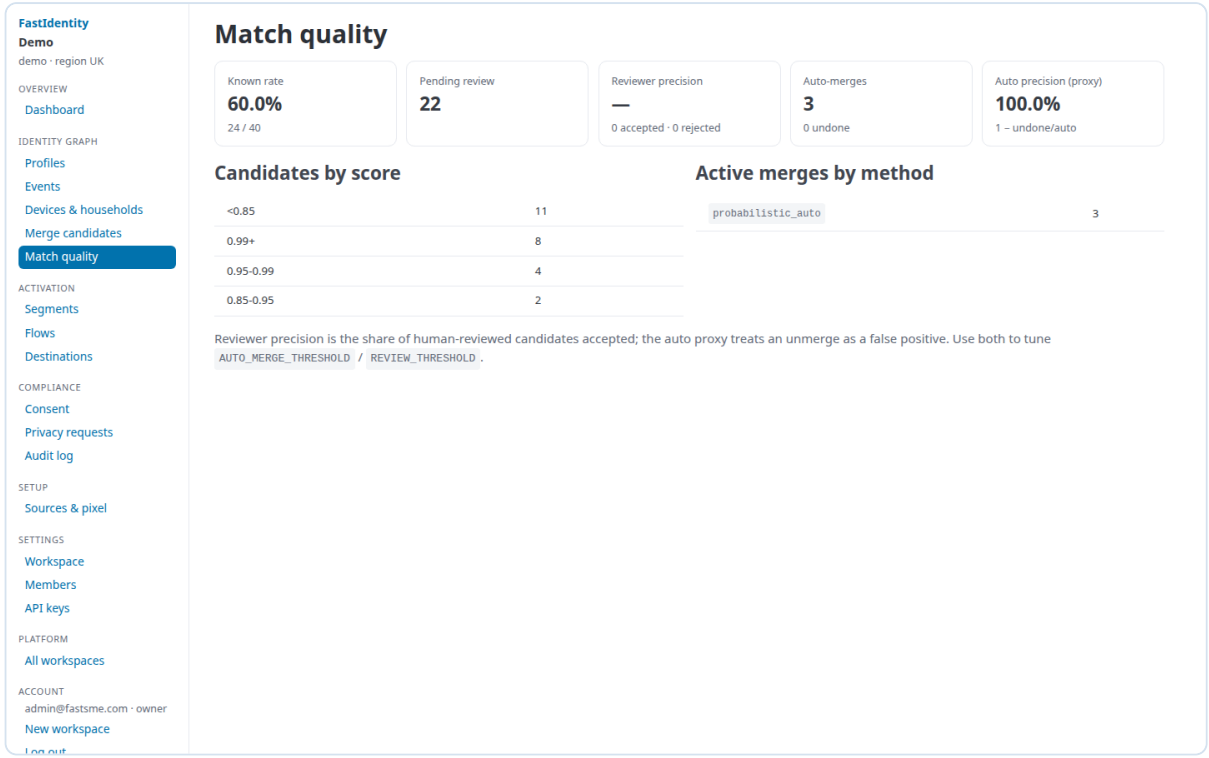


5. Identity graph — match quality

Match quality checks whether the matcher can be trusted.

- How auto, reviewed and deterministic merges break down.
- The score distribution of candidates.
- The **reviewer acceptance rate**: if staff reject many 0.9+ suggestions, raise `AUTO_MERGE_THRESHOLD`.
- The unmerge rate, which approximates the false-positive rate.

At larger scale, set `MATCH_ENGINE=splink` to fit the m/u parameters with EM on your own data instead of using the defaults.

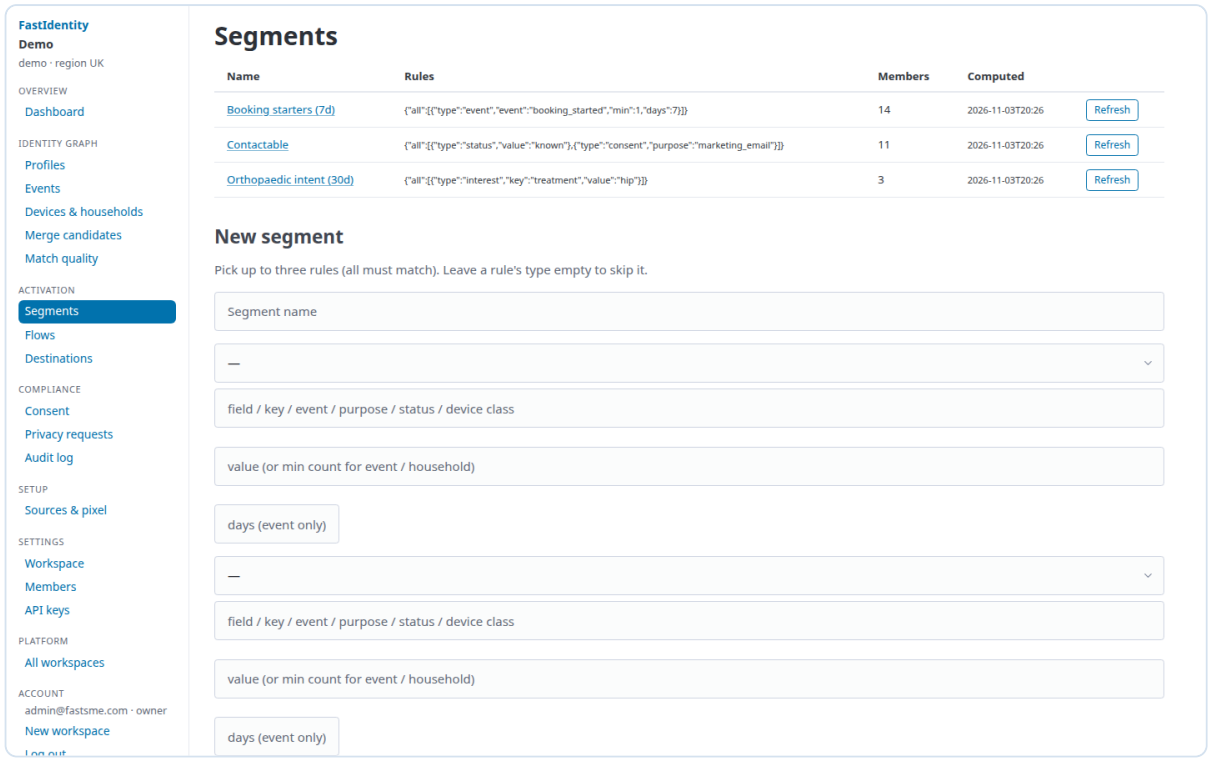


6. Activation — segments

Segments are saved audiences, defined as rules over traits, interests and recent events. For example: *viewed a knee-replacement package in the last 14 days, known, email-contactable*.

Rules can also use the **device type** a person used (e.g. *mobile only*) and **household** membership.

Segments feed flows and can be synced to a destination as a list. Membership is evaluated when a flow or sync runs, so segments never go stale.

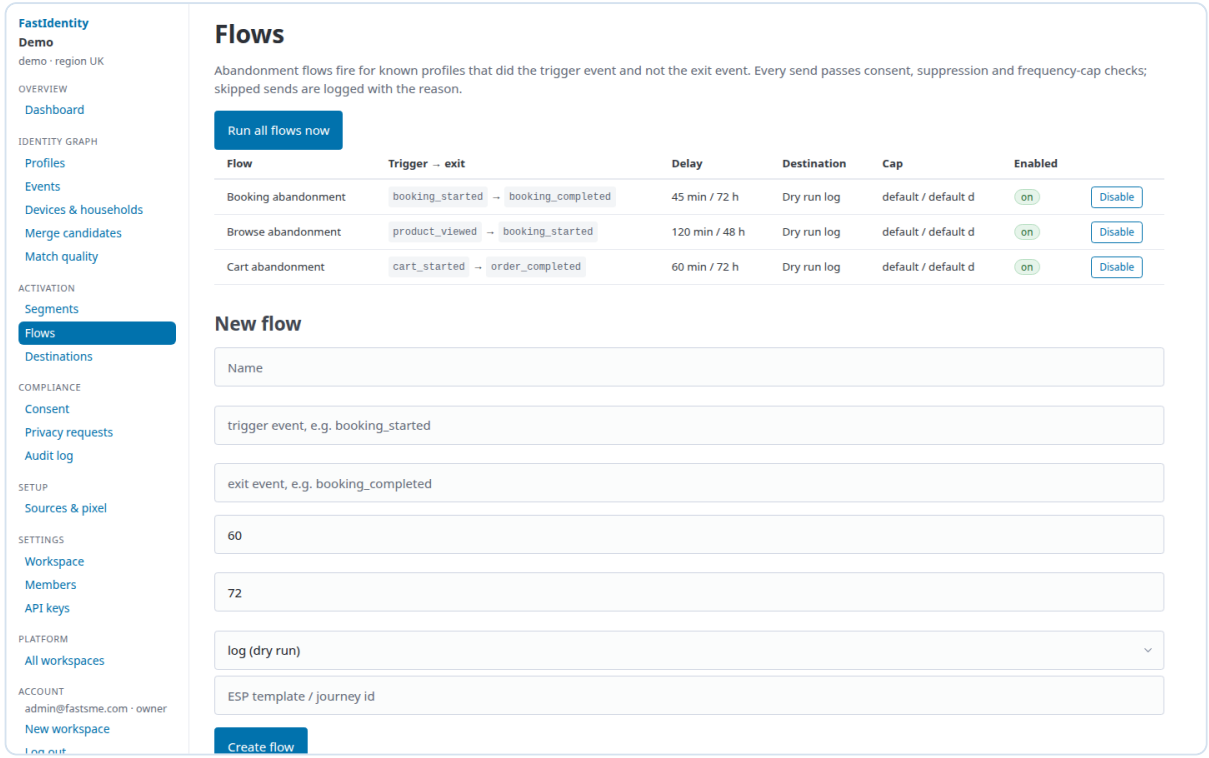


6. Activation — abandonment flows

A **flow** has a trigger event, an exit event, a delay and a lookback. Three templates ship with the product:

- **Browse abandonment**: viewed a treatment, then no enquiry within 24h.
- **Cart / quote abandonment**: started a quote, not completed.
- **Booking abandonment**: picked a date, did not book.

Before each send, the worker checks marketing consent, the hashed suppression list and the **frequency cap** (2 messages per 7 days by default). Skipped sends are recorded with the reason they were skipped.

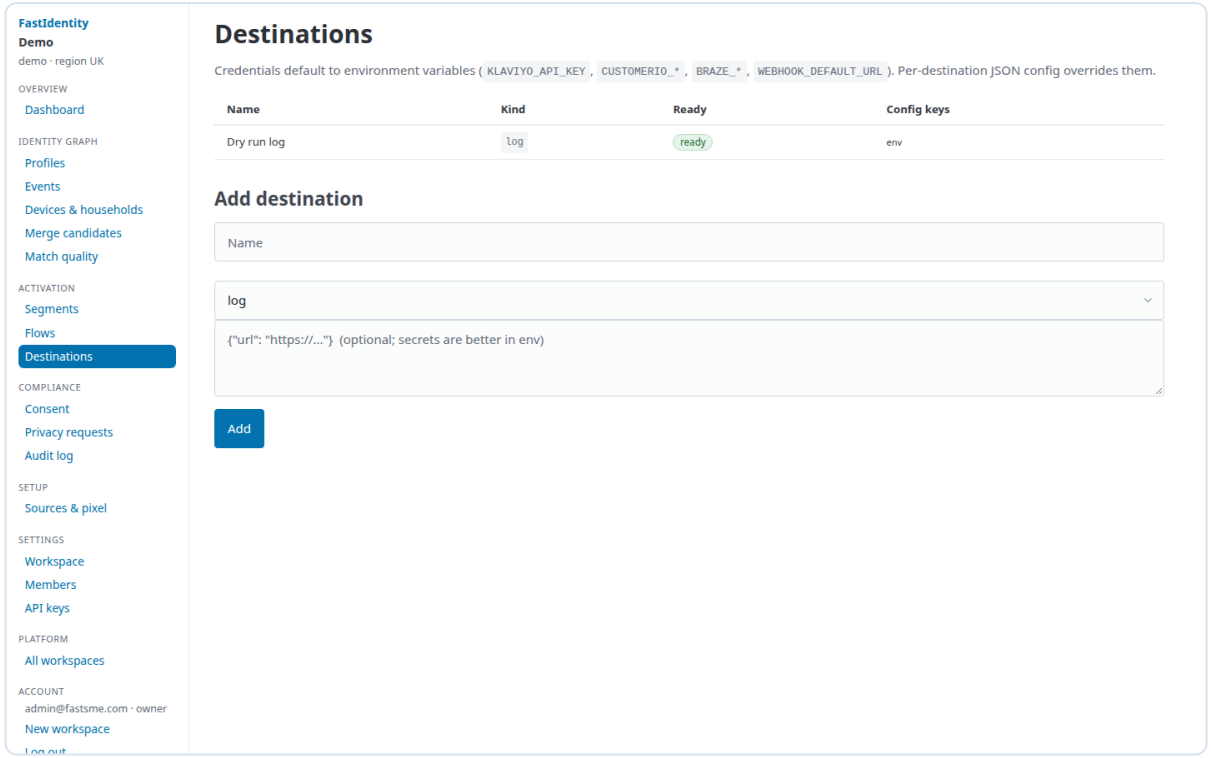


6. Activation — destinations

A **destination** is where activated events and profiles go.

- **Dry-run log**: the default. Shows exactly what *would* be sent.
- **Webhook**: an HMAC-signed POST to your own endpoint. This is the recommended way into MyMedicalGateway's Postmark sender.
- **Klaviyo**, **Customer.io** and **Braze** adapters, configured with API keys.

Details of each ESP are in `docs/esp_integration.md`.

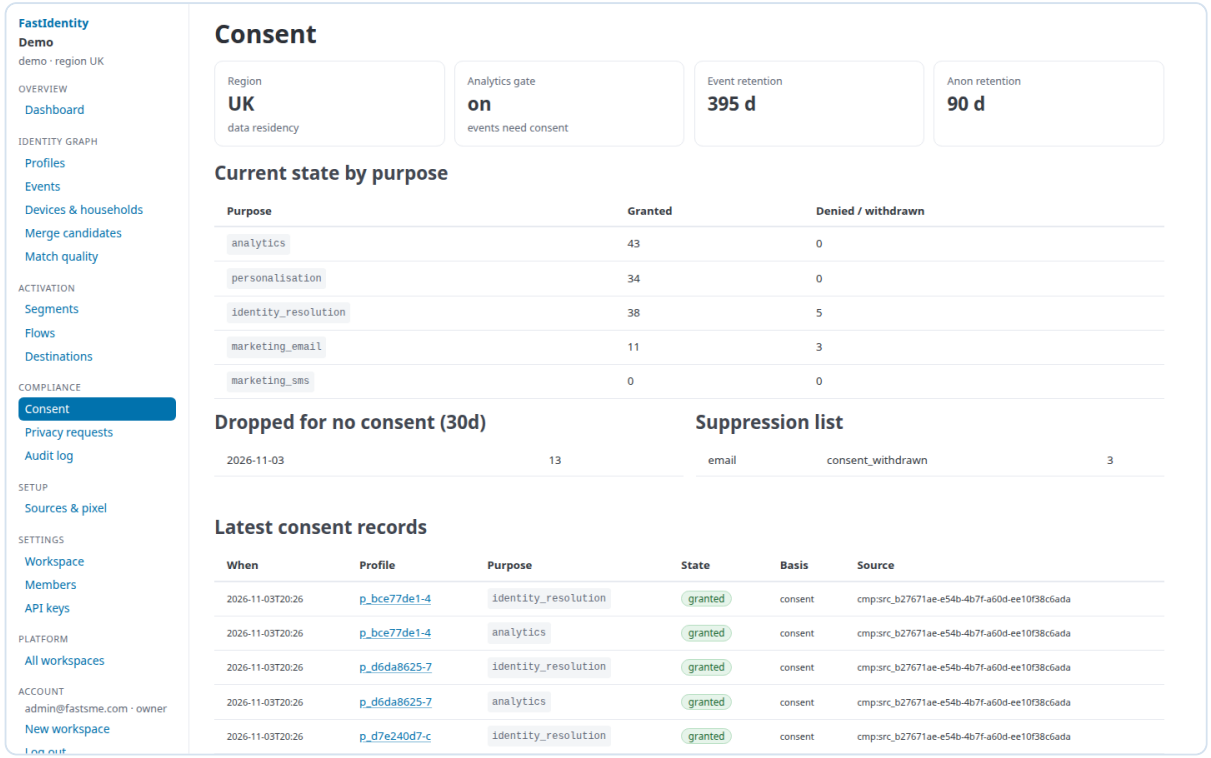


7. Compliance — consent ledger

Every consent change is **appended** to the ledger with a microsecond timestamp, its source, the legal basis and evidence (the CMP payload, the form wording and the page). Nothing is overwritten.

Purposes: `analytics`, `personalisation`, `identity_resolution`, `marketing_email`, `marketing_sms`.

A cookie banner never grants email consent. Marketing purposes come only from an explicit opt-in or the PECR soft opt-in. A withdrawal immediately adds the person's address to the suppression list. When two profiles merge, the most restrictive consent wins.

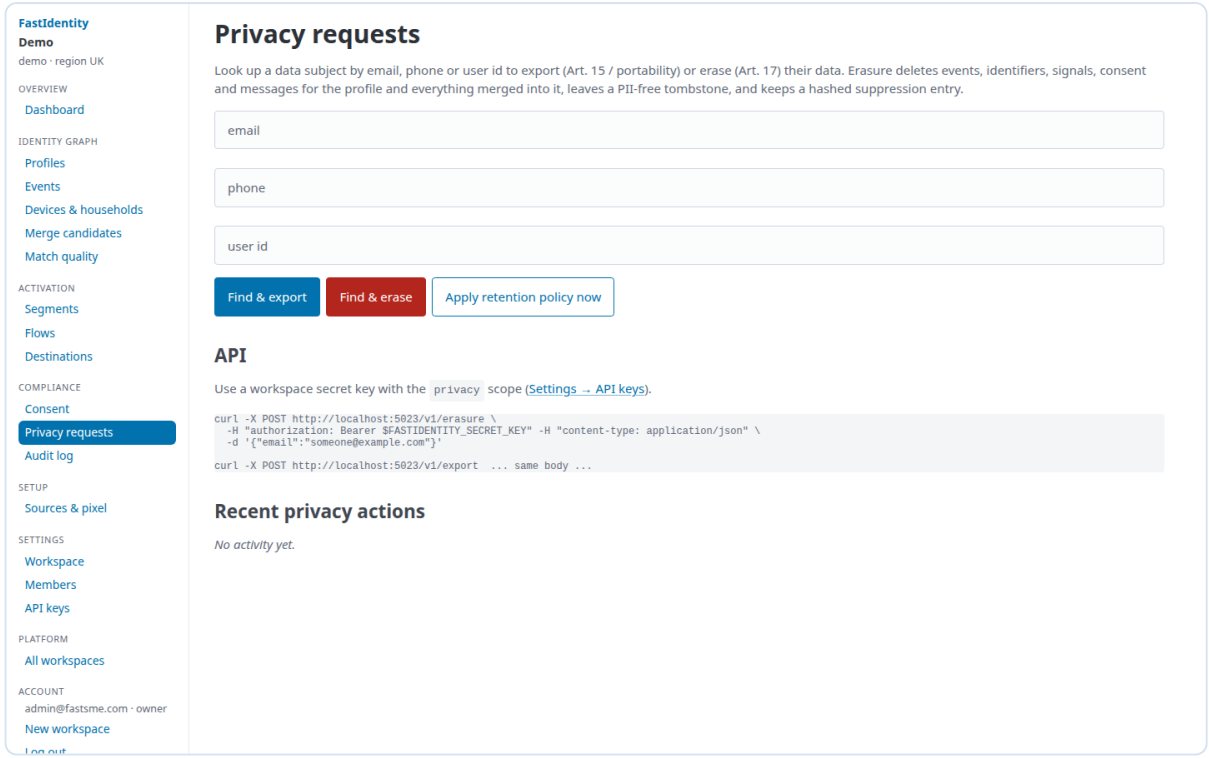


7. Compliance — privacy requests

Privacy requests handles GDPR Articles 15, 17 and 20.

- **Export (DSAR)**: a JSON file with the profile, identifiers, events, consent history and messages.
- **Erase**: deletes events, signals and identifiers and leaves a tombstone. The email or phone is kept only as a *hash* on the suppression list, so the person is never re-contacted.

The same operations are available server-to-server: `POST /v1/export` and `POST /v1/erasure`, authenticated with the admin token.

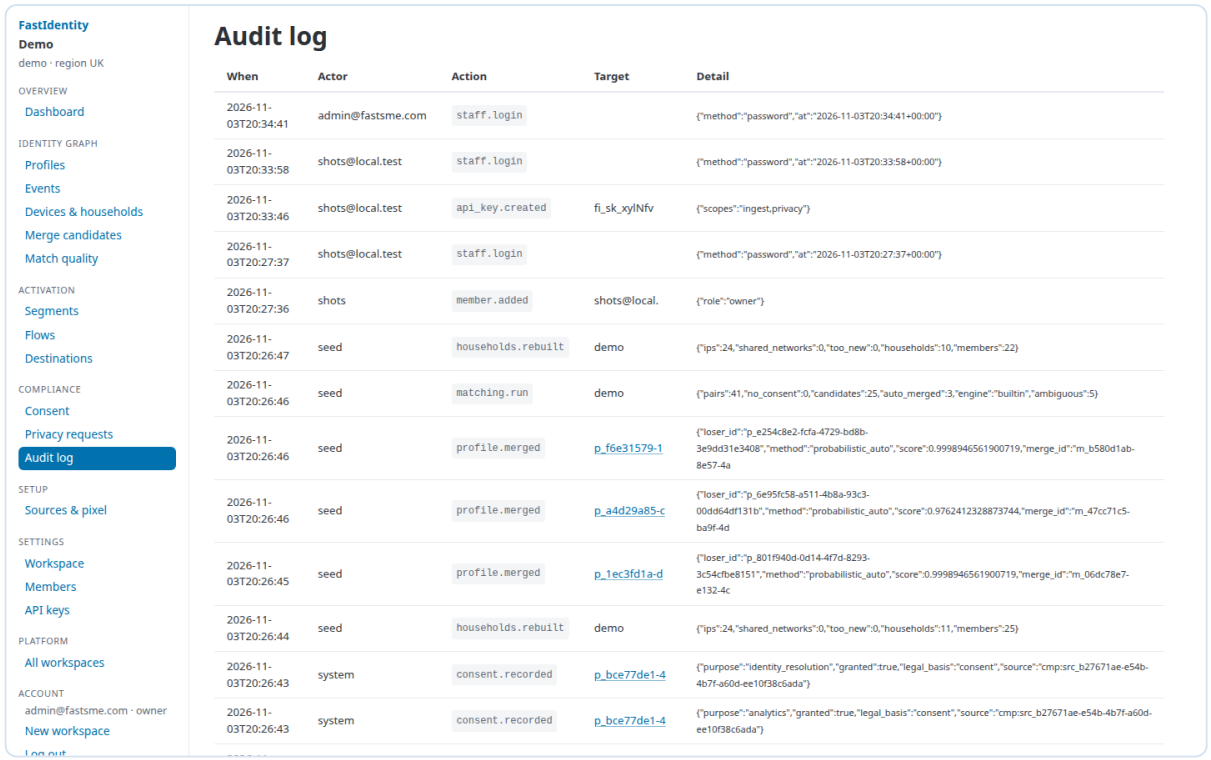


7. Compliance — audit log

The **audit log** is append-only. It records who did what, and when: staff logins, merges and unmerges with scores, consent changes, flow runs, erasures, and retention jobs.

It is the evidence trail for a DPIA, a regulator enquiry (ICO, or an EU supervisory authority), or a customer complaint.

Events older than 13 months (`EVENT_RETENTION_DAYS=395`) are purged automatically.



8. Setup — sources & pixel

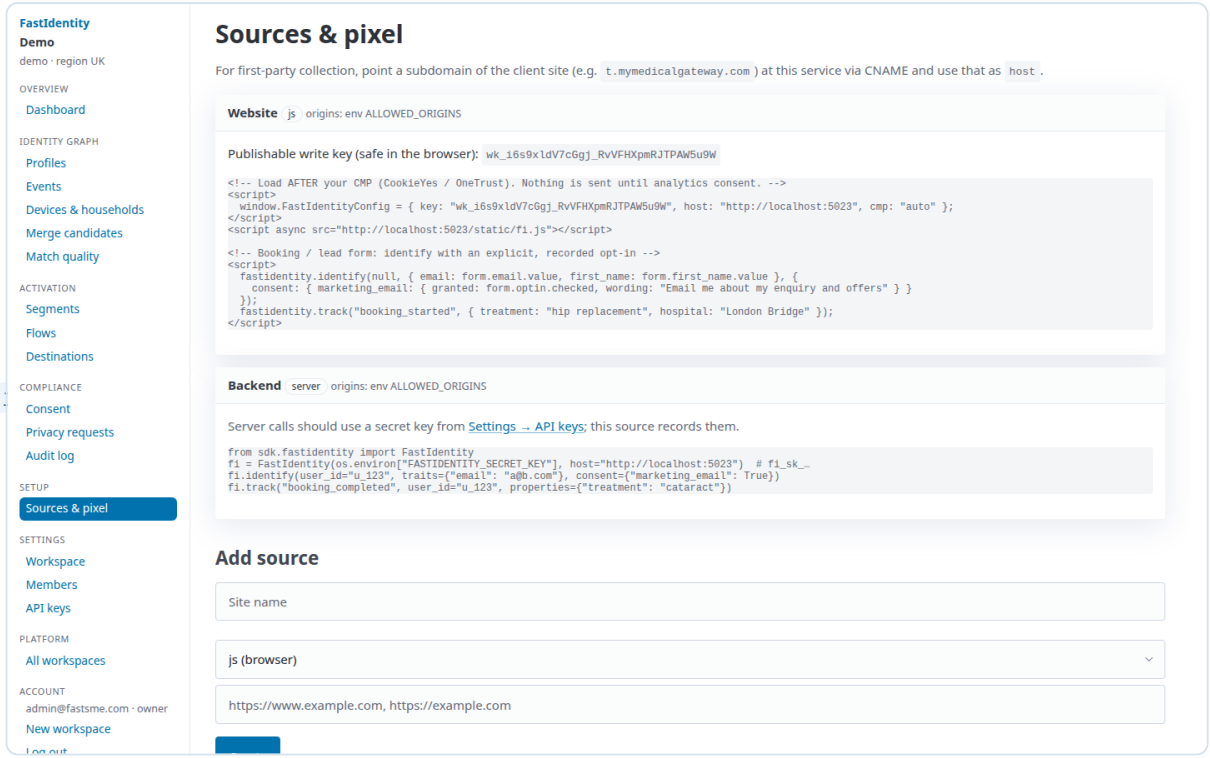
A **source** is anything that sends data, and each has its own write key.

- **Website (js)**: the browser pixel. Restricted to allowed origins.
- **Backend (server)**: server-side `identify` and `track` calls.

The **snippet** goes in `<head>` after the CMP script:

```
        <script src="https://identity.fastsme.com/stat
fi.js"
          data-write-key="WK_…" defer></script>
```

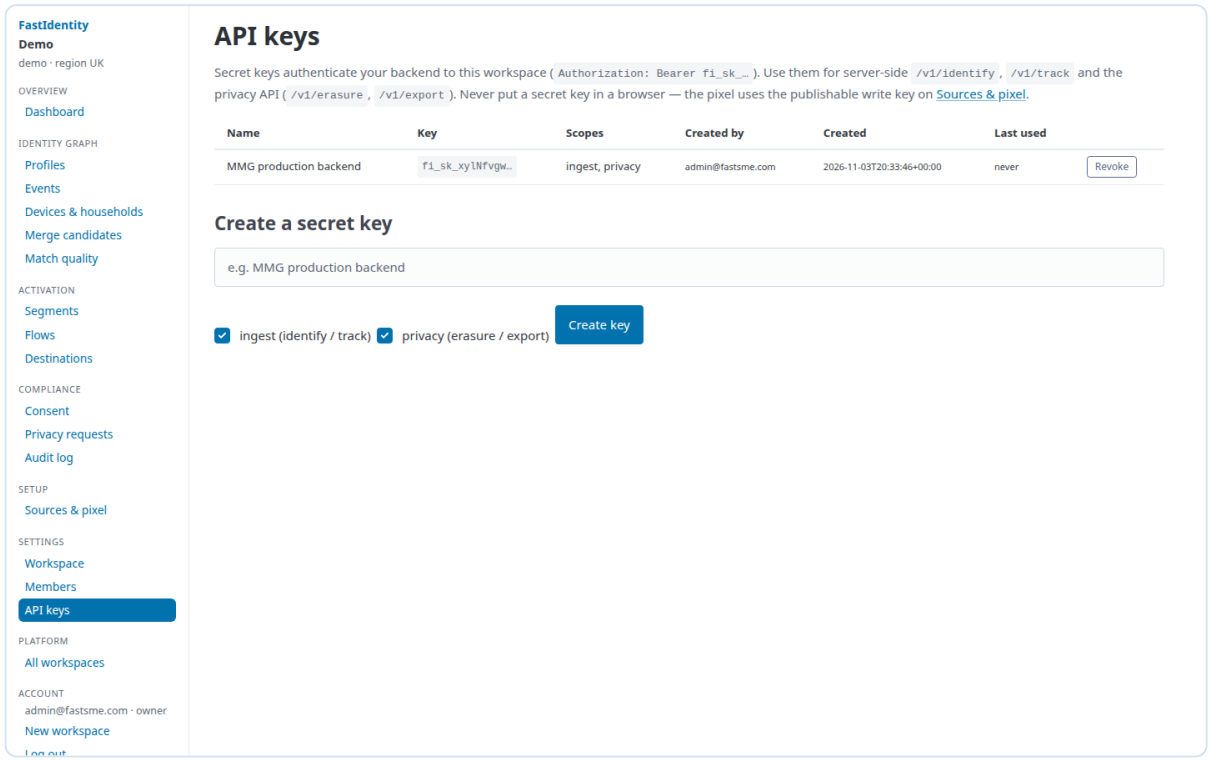
The pixel reads CookieYes, OneTrust or Consent Mode state. It sets its `_fi_aid` cookie only after analytics consent, and it sends device signals, including its first-party device ID (`_fi_did`), only with marketing consent. The device ID survives `fi.reset()` at logout, so a shared device is recognised, and it is deleted when consent is withdrawn.



8. Setup — workspaces, members & API keys

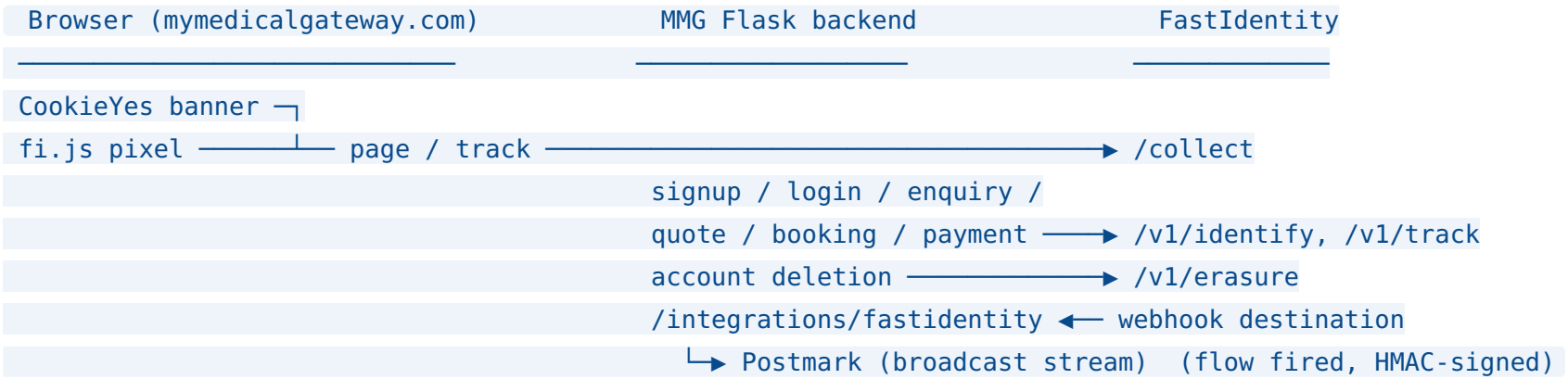
FastIdentity is **multi-tenant**. Each customer is a **workspace** with its own identity graph, consent ledger, sources and team. The same email in two workspaces is two different people.

- **Sign-up**: *Continue with Google* creates a workspace; you become its owner. Colleagues on the workspace's email domain join automatically.
- **Roles**: viewer, analyst (reviews merges, builds segments), admin (keys, members, exports, erasure) and owner (workspace settings, households).
- **Invites**: by email; the link is shown once.
- **Secret API keys** (`fi_sk_…`) authenticate a customer's backend. Each key is scoped to *ingest* and/or *privacy* (erasure and export), shown **once**, stored only as a hash, and revocable, with last-used time recorded.
- **Platform admins** (FastSME staff) can open any workspace. Each visit is audit-logged.



8. Setup — MyMedicalGateway: architecture

Context. mymedicalgateway.com is a server-rendered Flask app with Jinja templates. It uses **CookieYes** as its CMP, plus GTM, **Postmark** for email, Twilio for SMS, and an APScheduler worker (`mmg_scheduler`) that already sends time-based nudges.



Division of labour.

- **FastIdentity owns** recognising the visitor before they enquire, and the *browse* and *package-abandonment* flows.
- **MMG's scheduler keeps** the post-quote transactional nudges (diagnostics, balance reminders, callbacks).
- One rule across both: **MMG checks FastIdentity's suppression list before any marketing send**, and the frequency cap applies to both senders.

8. Setup — MyMedicalGateway: integration steps

1. **Pixel** (`mmg/templates/base.html`): add `fi.js` after the CookieYes and GTM snippets. The write key comes from `FASTIDENTITY_WRITE_KEY` through the `inject_now` context processor (`mmg/__init__.py`).
2. **Page context**: on treatment and hospital pages, call `fi.track("treatment_viewed", {treatment, hospital, country})`. The package page `/hospitals/<country>/<hospital>/<treatment>/` is the main abandonment trigger; its `…/start` POST counts as `quote_started`.
3. **Identify (server)**: call `/v1/identify` from the `user_registered` signal, after login (`/post-registration-handler`, Google OAuth), and on `register_interest`. Use `str(user.id)` as `userId`, not `fs_uniquifier`, because Flask-Security rotates that value. Send the `receive_email/sms` flags as consent with basis `consent`. From the mobile app, forward the vendor device ID as `context.device.id` (never the advertising ID).
4. **Conversion events (server)**: `enquiry_submitted`, `quote_created` (`/treatments/terms`), `booking_submitted` (`/book/submit`), `deposit_paid` (from the payment webhook). These are the flows' *exit* events.
5. **Logout**: call `fi.reset()` on the logged-out page.
6. **Consent capture**: add an explicit, unticked marketing-email opt-in to the patient enquiry and registration forms. Today these flags can only be edited in the provider app.
7. **Erasure and export**: add a patient account-deletion route to `user_bp`. It should call `/v1/erasure` and `/v1/export`; MMG has no such route today.
8. **Activation**: create a webhook destination pointing at `/integrations/fastidentity/webhook`. MMG verifies the HMAC and sends through Postmark's **broadcast** message stream, not the transactional one.
9. **Config**: add `FASTIDENTITY_URL`, `FASTIDENTITY_WRITE_KEY` (the publishable pixel key), `FASTIDENTITY_SECRET_KEY` and `FASTIDENTITY_WEBHOOK_SECRET` to `.envs.example/*/patient.env`. The secret key comes from *Settings → API keys* in the `mymedicalgateway` workspace with ingest and privacy scopes; it is shown once. Mark them skippable in `config/settings/base.py` so local development works without them.

8. Setup — MyMedicalGateway: health data & go-live

Special-category data. Interest in a specific treatment, linked to a named person, is *data concerning health* under GDPR Art. 9 and UK GDPR. For MMG this means:

- **Explicit consent** for treatment-specific marketing. Abandonment emails go only to people with `marketing_email` consent captured on a form, never under the soft opt-in.
- Email copy refers to the *destination or hospital*, never to the *diagnosis*. Subject lines must never reveal the treatment.
- A **DPIA** is completed before go-live, and FastIdentity is added to MMG's record of processing activities and privacy notice as a processor.
- **Households stay off** in the MMG workspace. A shared home IP must never route one patient's treatment interest to a partner or parent. Device-level linking is fine under `identity_resolution` consent.
- Data stays in the UK region (`DATA_REGION=uk`). The ESP receives the minimum: an email address and a flow ID.

Go-live checklist.

- [ ] Pixel live on beta; consent drops visible on the dashboard
- [ ] identify / track events arriving from the backend; deterministic merges confirmed
- [ ] Webhook destination tested in dry-run, then a staff-only segment
- [ ] Erasure round-trip tested (MMG deletion → FastIdentity tombstone → suppression)
- [ ] DPIA signed off; privacy notice updated
- [ ] Production secret key created in the MMG workspace and stored in MMG's secret store; staff keys revoked

Tracking issue: mymedicalgateway/mmg#1420 (milestone: 4 November 2026).

9. Operating playbook

Daily (5 minutes).

- Dashboard: consent drops and pending merges.
- Clear the merge-candidate queue.

Weekly.

- Match quality: if the reviewer acceptance rate on 0.9+ candidates is above 95%, consider lowering the review threshold; if it is below 80%, raise `AUTO_MERGE_THRESHOLD`.
- Devices & households: watch the shared-device list. A device shared by many profiles is usually a kiosk or reception PC; check that its candidates are being rejected.
- Flows: check messages sent against messages skipped. A high “no_consent” count means the opt-in on forms needs work.

Monthly.

- Audit log spot-check.
- Confirm retention jobs ran.
- Review DSAR and erasure turnaround (the target is under 30 days; ours is instant).

Deploy. Push to `main` and the GitHub webhook triggers a Coolify build. The new version takes over only after the `/health` check passes. Manual operations: `python scripts/coolify.py status | env --sync --yes | deploy --yes`.

Never rotate `IDENTITY_HASH_SECRET` casually. It permanently breaks probabilistic linkage to all previously hashed signals.

Appendix — Look-alike modelling

A **look-alike** is two jobs, and FastIdentity owns only the first: it builds a clean, consented **seed** and ranks the first-party base by resemblance to it; the ad platform (Meta Lookalike, Google Similar) does the population-level **expansion** from its own graph. Only profiles already in the identity graph are ever scored — strangers are never modelled.

`domain.lookalike` is a small, auditable scikit-learn baseline:

- **Features (first-party only).** Status, recency/tenure, event volume and mix (incl. `product_viewed`, `booking_started`, `order_completed`), resolved interests (`int=treatment:hip replacement`), device classes and household membership. Raw PII, fingerprints and IPs are never features. **Consent is a gate, not a feature.**
- **Model.** The seed is the positive class; a random sample of the rest are treated as negatives (PU-learning baseline). A logistic-regression pipeline (`DictVectorizer` → `MaxAbsScaler` → `LogisticRegression`, balanced classes) yields a calibrated, inspectable score in `[0, 1]`; `top_features` explains every run.
- **Output.** Non-seed profiles ranked by score with status + consent flags — ready to filter by consent, hash and sync as a seed, or to prioritise owned-channel outreach. `lookalike_for_segment()` seeds straight from a saved segment (bookers, high-intent, high-value specialty).
- **Guardrails.** A minimum seed size (default 25; platforms want ~100+ matched); seed on **behaviour/stage, never diagnosis**; exclude converters/opt-outs; re-score on a schedule. A dependency-free cosine-similarity `builtin` engine keeps the feature working where scikit-learn is not installed.

Full detail — features, the PU method, outputs, configuration and the compliance rules — is in **docs/lookalike_methodology.md** (PDF).